VALIDATION REPORT
OPERATIONAL QUALIFICATION
CORRELATION ANALYSIS MODULE

JR Validated Environment — Correlation Analysis Module v1.0

Field	Value
Document Number	JR-VR-CORR-001
Title	Validation Report — Operational Qualification, Correlation Analysis Module
Validation Plan	JR-VP-CORR-001 v1.0
System	JR Validated Environment — Correlation Analysis Module
Module Version	1.0
Document Version	1.1
Execution Date	2026-05-18
OQ Result	PASS — 50 / 50 tests passed, 0 failures, 0 deviations
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-03-20	Joep Rous	Initial OQ execution report. All 45 test cases passed. No deviations.
1.1	2026-05-18	Joep Rous	Regenerated from OQ evidence file. 50 test cases executed (50 passed, 0 failed). Includes numeric correctness TCs (P-012..013, R-012..014) added in v2.5.0.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-05-18
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document reports the execution and outcome of the Operational Qualification (OQ) test suite for the JR Validated Environment Correlation Analysis Module, version 1.0. It provides objective evidence that all Corr scripts perform as specified in the OQ Validation Plan JR-VP-CORR-001 v1.0 under the defined test conditions.

This report records the test environment, execution results for all 50 test cases, and the formal OQ conclusion. Results are derived directly from the OQ evidence file produced by admin_corr_oq.

2. Scope

2.1 In Scope

- All 4 R scripts in repos/corr/R/ (jrc_corr_pearson, jrc_corr_spearman, jrc_corr_regression, jrc_corr_passing_bablok)
- Correct computation of correlation coefficients, regression coefficients, CI bounds, and p-values for valid inputs
- Numeric correctness assertions for exact linear data (Pearson r, OLS slope, intercept, R-squared)
- Correct rejection of invalid inputs with informative error messages
- Bypass-protection verification (RENV_PATHS_ROOT check)
- PNG output to ~/Downloads/

2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by JR-IQ-001)
- Performance qualification (PQ) with real production data
- Validation of R interpreter and third-party packages (ggplot2, grid)
- Statistical validation of the underlying methods (Pearson, Spearman, OLS, Passing-Bablok) — method references are cited in JR-VP-CORR-001 Section 3

3. Reference Documents

Document ID	Title	Location
JR-VP-CORR-001 v1.0	Validation Plan — OQ, Correlation Analysis Module	repos/corr/docs/corr_validation_plan.pdf
JR-VP-002 v1.0	Validation Plan — OQ, Community Script Suite v1.1.0	docs/statistics_validation_plan.pdf
JR-IQ-001	IQ Execution Evidence	docs/core_IQ_validation_plan.pdf
OQ Evidence	Corr OQ Execution Evidence (pytest output + header)	corr_oq_execution_20260518T172631.txt

4. Test Environment

Item	Value
Execution date / time	2026-05-18T17:26:31
Host	Joeps-MacBook-Air.local

Operating system	15.7.3
R version	4.6.0
Python version	3.11.9
pytest version	pytest 8.3.4
OQ virtual environment	~/.venvs/<PROJECT_ID>_oq/ (reused from core OQ suite)
Script entry point	bin/jrrun (sets RENV_PATHS_ROOT, loads renv library)
Test runner	repos/corr/admin_corr_oq
Test directory	repos/corr/oq/
Test data directory	repos/corr/oq/data/ (7 committed synthetic CSV files)
OQ evidence file	corr_oq_execution_20260518T172631.txt

5. Test Data

All test data files are synthetic CSV files committed to repos/corr/oq/data/ and included in the project integrity check (admin/project_integrity.sha256). No real patient or production data is used.

File	Content	Used by
corr_linear.csv	n=10, y≈2x+1 with small noise; r≈0.999 strong positive correlation	TC-CORR-P-001..005, TC-CORR-S-001..005, TC-CORR-R-001..005
corr_exact_linear.csv	n=10, y=2x+1 exactly (x=1..10); r=1.000, slope=2.000, intercept=1.000, R^2=1.000	TC-CORR-P-012..013, TC-CORR-R-012..014
corr_negative.csv	n=10, y≈−2x+22; r≈−0.999 strong negative correlation	TC-CORR-P-010, TC-CORR-S-010
corr_uncorrelated.csv	n=10, irregular y values; |r| < 0.5, not significant	Supplementary
corr_method_comp.csv	n=9, y≈1.1x+2 with small noise; method comparison data	TC-CORR-PB-001..006, TC-CORR-PB-011
corr_small.csv	n=2 pairs only — triggers minimum-n error	TC-CORR-P-008, TC-CORR-S-008, TC-CORR-R-008, TC-CORR-PB-009
corr_nonnumeric.csv	y column contains letters A, B, C — triggers non-numeric error	TC-CORR-P-009, TC-CORR-S-009, TC-CORR-R-009, TC-CORR-PB-010

6. Test Execution Results

6.1 Summary

Item	Value
Total test cases	50
Passed	50
Failed	0
Errors	0
OQ result	**PASS**

6.2 Results by Test File

Test file	Script covered	Tests	Result
test_corr_pearson.py	jrc_corr_pearson	13	**13 / 13 PASS**

test_corr_spearman.py	jrc_corr_spearman	11	**11 / 11 PASS**
test_corr_regression.py	jrc_corr_regression	14	**14 / 14 PASS**
test_corr_passing_bablok.py	jrc_corr_passing_bablok	12	**12 / 12 PASS**

6.3 Individual Test Case Results

All 50 test cases were executed. The full pytest output, including individual test IDs, pass/fail status, and timing, is recorded in the evidence file. Per-test results read from that file are reproduced in the table below for traceability.

Test Case ID	Description	Expected	Result
TC-CORR-P-001	Happy path — exit 0, 'Pearson' in output	Exit 0, 'Pearson' and 'Pearson r:' present	**PASS**
TC-CORR-P-002	Pearson r label present	Exit 0, 'Pearson r:' in output	**PASS**
TC-CORR-P-003	CI values present in output	Exit 0, '[' and ']' in output	**PASS**
TC-CORR-P-004	p-value present in output	Exit 0, 'p-value' in output	**PASS**
TC-CORR-P-005	PNG written to ~/Downloads/	PNG present, recent mtime	**PASS**
TC-CORR-P-006	No arguments → usage message	Exit ≠ 0, 'Usage' in output	**PASS**
TC-CORR-P-007	File not found	Exit ≠ 0, 'not found' in output	**PASS**
TC-CORR-P-008	n < 3 (corr_small.csv) → error	Exit ≠ 0, '3' in output	**PASS**
TC-CORR-P-009	Non-numeric column → error	Exit ≠ 0	**PASS**
TC-CORR-P-010	Negative correlation → r < 0 in output	Exit 0, '-0.' in output	**PASS**
TC-CORR-P-011	Bypass protection — direct Rscript call	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	**PASS**
TC-CORR-P-012	Pearson r = 1.000 ± 0.001 (corr_exact_linear.csv)	Exit 0; Pearson r: 1.000 ± 0.001	**PASS**
TC-CORR-P-013	p-value significant for exact linear data	Exit 0; p-value < 0.001, 'significant' in output	**PASS**
TC-CORR-S-001	Happy path — exit 0, 'Spearman' in output	Exit 0, 'Spearman' present	**PASS**
TC-CORR-S-002	Spearman rho label present	Exit 0, 'Spearman rho:' in output	**PASS**
TC-CORR-S-003	p-value present in output	Exit 0, 'p-value' in output	**PASS**
TC-CORR-S-004	CI note present	Exit 0, 'Confidence interval' note present	**PASS**
TC-CORR-S-005	PNG written to ~/Downloads/	PNG present, recent mtime	**PASS**
TC-CORR-S-006	No arguments → usage message	Exit ≠ 0, 'Usage' in output	**PASS**
TC-CORR-S-007	File not found	Exit ≠ 0, 'not found' in output	**PASS**
TC-CORR-S-008	n < 3 → error	Exit ≠ 0	**PASS**
TC-CORR-S-009	Non-numeric column → error	Exit ≠ 0	**PASS**
TC-CORR-S-010	Negative correlation → rho < 0 in output	Exit 0, '-0.' in output	**PASS**
TC-CORR-S-011	Bypass protection — direct Rscript call	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	**PASS**
TC-CORR-R-001	Happy path — exit 0, 'Regression' in output	Exit 0, 'Regression' present	**PASS**
TC-CORR-R-002	Slope label present	Exit 0, 'Slope' in output	**PASS**
TC-CORR-R-003	Intercept label present	Exit 0, 'Intercept' in output	**PASS**
TC-CORR-R-004	R-squared present in output	Exit 0, 'R-squared' in output	**PASS**
TC-CORR-R-005	PNG written to ~/Downloads/	PNG present, recent mtime	**PASS**
TC-CORR-R-006	No arguments → usage message	Exit ≠ 0, 'Usage' in output	**PASS**
TC-CORR-R-007	File not found	Exit ≠ 0, 'not found' in output	**PASS**
TC-CORR-R-008	n < 3 → error	Exit ≠ 0	**PASS**
TC-CORR-R-009	Non-numeric column → error	Exit ≠ 0	**PASS**

TC-CORR-R-010	Slope ≈ 2.0 for corr_linear.csv	Exit 0; '1.9' or '2.0' in output	**PASS**
TC-CORR-R-011	Bypass protection — direct Rscript call	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	**PASS**
TC-CORR-R-012	OLS slope = 2.000 ± 0.001 (corr_exact_linear.csv)	Exit 0; Slope (b1): 2.000 ± 0.001	**PASS**
TC-CORR-R-013	OLS intercept = 1.000 ± 0.001 (corr_exact_linear.csv)	Exit 0; Intercept (b0): 1.000 ± 0.001	**PASS**
TC-CORR-R-014	R-squared = 1.000 ± 0.001 (corr_exact_linear.csv)	Exit 0; R-squared: 1.000 ± 0.001	**PASS**
TC-CORR-PB-001	Happy path — exit 0, 'Passing-Bablok' in output	Exit 0, 'Passing-Bablok' present	**PASS**
TC-CORR-PB-002	Slope label present	Exit 0, 'Slope' in output	**PASS**
TC-CORR-PB-003	Intercept label present	Exit 0, 'Intercept' in output	**PASS**
TC-CORR-PB-004	Cusum test result present	Exit 0, 'Cusum' in output	**PASS**
TC-CORR-PB-005	Proportionality test present	Exit 0, slope=1 test in output	**PASS**
TC-CORR-PB-006	PNG written to ~/Downloads/	PNG present, recent mtime	**PASS**
TC-CORR-PB-007	No arguments → usage message	Exit ≠ 0, 'Usage' in output	**PASS**
TC-CORR-PB-008	File not found	Exit ≠ 0, 'not found' in output	**PASS**
TC-CORR-PB-009	n < 3 → error	Exit ≠ 0	**PASS**
TC-CORR-PB-010	Non-numeric column → error	Exit ≠ 0	**PASS**
TC-CORR-PB-011	Slope ≈ 1.1 for corr_method_comp.csv	Exit 0; '1.1' in output	**PASS**
TC-CORR-PB-012	Bypass protection — direct Rscript call	Exit ≠ 0, 'RENV_PATHS_ROOT' in output	**PASS**

7. Deviations from OQ Plan (JR-VP-CORR-001 v1.0)

No deviations from the approved OQ plan were identified during this OQ execution.

ID	Description	Resolution	Status
—	None	—	N/A

8. Requirements Traceability Matrix

The table below maps each User Requirement from JR-VP-CORR-001 to the test cases executed and their result.

UR	Requirement (summary)	Test Cases	Result
UR-CORR-001	Pearson r, CI (Fisher z), t-stat, p-value	TC-CORR-P-001..004, P-012	**PASS**
UR-CORR-002	Pearson correlation classification (weak/moderate/strong, direction)	TC-CORR-P-001, TC-CORR-P-010	**PASS**
UR-CORR-003	Pearson scatter PNG to ~/Downloads/	TC-CORR-P-005	**PASS**
UR-CORR-004	Spearman rho, S statistic, p-value	TC-CORR-S-001..003	**PASS**
UR-CORR-005	Spearman classification; CI note displayed	TC-CORR-S-001, TC-CORR-S-004, TC-CORR-S-010	**PASS**
UR-CORR-006	Spearman rank scatter PNG to ~/Downloads/	TC-CORR-S-005	**PASS**
UR-CORR-007	OLS regression: slope, intercept, R^2, CI, F-stat, p-value	TC-CORR-R-001..004, R-010, R-012..014	**PASS**
UR-CORR-008	Regression two-panel PNG (scatter + residuals) to ~/Downloads/	TC-CORR-R-005	**PASS**
UR-CORR-009	Passing-Bablok slope, intercept, CI (native algorithm)	TC-CORR-PB-001..003, TC-CORR-PB-011	**PASS**

UR-CORR-010	Proportionality test (slope=1) and bias test (intercept=0)	TC-CORR-PB-005	**PASS**
UR-CORR-011	Cusum linearity test	TC-CORR-PB-004	**PASS**
UR-CORR-012	Passing-Bablok PNG with identity line to ~/Downloads/	TC-CORR-PB-006	**PASS**
UR-CORR-013	--xcol / --ycol column selection	TC-CORR-P-001, TC-CORR-S-001, TC-CORR-R-001, TC-CORR-PB-001	**PASS**
UR-CORR-014	All scripts — RENV_PATHS_ROOT bypass protection	TC-CORR-P-011, TC-CORR-S-011, TC-CORR-R-011, TC-CORR-PB-012	**PASS**
UR-CORR-015	All scripts — no-arguments usage message	TC-CORR-P-006, TC-CORR-S-006, TC-CORR-R-006, TC-CORR-PB-007	**PASS**
UR-CORR-016	All scripts — file not found error	TC-CORR-P-007, TC-CORR-S-007, TC-CORR-R-007, TC-CORR-PB-008	**PASS**
UR-CORR-017	All scripts — reject n < 3	TC-CORR-P-008, TC-CORR-S-008, TC-CORR-R-008, TC-CORR-PB-009	**PASS**
UR-CORR-018	All scripts — reject non-numeric column	TC-CORR-P-009, TC-CORR-S-009, TC-CORR-R-009, TC-CORR-PB-010	**PASS**

9. OQ Conclusion

The Operational Qualification of the JR Validated Environment Correlation Analysis Module v1.0 is complete. All 50 test cases defined in JR-VP-CORR-001 were executed on 2026-05-18 and passed with no failures, no errors, and no deviations.

All 18 user requirements (UR-CORR-001 through UR-CORR-018) are covered by at least one passing test case. The OQ acceptance criteria stated in JR-VP-CORR-001 Section 10 are fully met.

The Correlation Analysis module is hereby declared OPERATIONALLY QUALIFIED and released for use in design verification activities in accordance with the JR Validated Environment quality system.

OQ Evidence file:

/Users/joeprous/.jrscript/MyProject/validation/corr_oq_execution_20260518T172631.txt